💻 Computer Science Roadmap (Week-wise Structured Plan)

🧠 PHASE 1: Programming Fundamentals

📅 Weeks 1–6

📝 Week 1: Getting Started with Programming

- Install VS Code, Git
- Learn:
  - What is programming
  - Syntax basics (Python recommended)
- Topics:
  - Variables
  - Data types (int, float, string, boolean)

👉 Practice:

- 10–15 basic problems (input/output)

📝 Week 2: Control Flow

- Learn:
  - if / else
  - nested conditions
- Logic building

👉 Practice:

- Even/odd, grading system, calculator

📝 Week 3: Loops

- Learn:

    - for loop
    - while loop

👉 Practice:

- Patterns (stars)
- Sum of numbers
- Factorial

📓 Week 4: Functions

- Learn:
    - Function creation
    - Parameters & return values

👉 Practice:

- Modular coding problems

📓 Week 5: Data Structures Intro

- Lists / Arrays
- Strings

👉 Practice:

- Reverse string
- Palindrome check

📓 Week 6: OOP Basics

- Classes & Objects
- Basic concepts

👉 Mini Project:

- Student management system

⍰ PHASE 2: Data Structures & Algorithms

📅 Weeks 7–14

📘 Week 7: Arrays Deep Dive

- Traversing
- Searching

👉 Practice:

- 10–15 array problems

📘 Week 8: Strings Advanced

- Manipulation
- Pattern matching

📘 Week 9: Linked Lists

- Insert, delete, traverse

📘 Week 10: Stack & Queue

- LIFO / FIFO
- Applications

📘 Week 11: HashMaps

- Key-value storage
- Fast lookup

Week 12: Trees

- Binary trees
- Traversals (inorder, preorder)

Week 13: Graphs

- BFS / DFS
- Graph representation

Week 14: Algorithms

- Sorting:
  - Merge sort
  - Quick sort
- Binary search

PHASE 3: Core CS Subjects

Weeks 15–20

Week 15: Operating Systems

- Processes & Threads
- Memory basics

Week 16: OS Advanced

- Scheduling
- Deadlocks

📓 Week 17: DBMS Basics

- SQL
- CRUD operations

📓 Week 18: DBMS Advanced

- Joins
- Indexing

📓 Week 19: Computer Networks

- HTTP/HTTPS
- TCP/IP

📓 Week 20: System Design Intro

- Client-server
- APIs

🧩 PHASE 4: Development (Build Skills)

📅 Weeks 21–32

📓 Week 21–22: HTML & CSS

- Structure pages
- Styling

👉 Project:

- Personal portfolio

📓 Week 23–24: JavaScript Basics

- Variables
- DOM manipulation

📓 Week 25–26: Advanced JavaScript

- Events
- Async JS

📓 Week 27–28: React

- Components
- State & props

👉 Project:

- To-do app

📓 Week 29–30: Backend Development

- Node.js / Django
- APIs

📓 Week 31–32: Database Integration

- Connect backend + DB
- CRUD app

👉 Project:

- Blog website

🧩 PHASE 5: Advanced Tools

📅 Weeks 33–40

📘 Week 33–34: Git & GitHub

- Version control
- Collaboration

📘 Week 35–36: Docker

- Containers
- Deployment basics

📘 Week 37–38: Cloud (AWS Basics)

- Hosting apps
- Deployment

📘 Week 39–40: CI/CD

- Automation
- GitHub Actions

🟣 PHASE 6: Projects (Portfolio Building)

📅 Weeks 41–48

📘 Week 41–42: Beginner Projects

- Calculator
- Notes app

📓 Week 43–44: Intermediate Projects

- Blog app
- Chat app

📓 Week 45–46: Advanced Projects

- E-commerce site
- Authentication system

📓 Week 47–48: Final Project

- Full-stack SaaS project

🧠 PHASE 7: Job Preparation

📅 Weeks 49–52

📓 Week 49: Resume + LinkedIn

- Build strong profile

📓 Week 50: DSA Revision

- Solve daily problems

📓 Week 51: Mock Interviews

- Practice interviews

📝 Week 52: Applications

- Apply for internships/jobs

⏰ Weekly Routine (Important)

Every week:

- 5 days learning
- 2 days practice/projects

Daily:

- 2–4 hours minimum

🎯 Final Outcome (After 1 Year)

By week 52, you will have:

✅ Strong coding skills
✅ DSA knowledge
✅ 4–6 real projects
✅ GitHub portfolio
✅ Internship/job readiness